GEK-83774

Instructions

Care and Storage of Magnetic Tapes

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CARE AND STORAGE OF MAGNETIC TAPES

Most magnetic tape manufacturers estimate that data retention on magnetic tapes to be much greater than fifty years with no noticeable loss in signal. In fact, BASF stated that a paper-backed tape is located in archives in Germany which was recorded in 1932 and is still playable.

However, many tapes fail before their life expectancy, and the reason for this is, in most all cases, inadequate care of the magnetic tapes. There have been a few instances where dropouts are blamed on a minute section of tape losing its oxide coating, or on imperfections in the coating, but almost every other failure is due to poor handling by operators or lack of precautions.

CONTAMINATION

Improper care of magnetic tape usually results in contamination. Dust, food, liquid, paper particles, lint, or even loose oxide particles on the tape will cause it to lift off the head to pass over these contaminates which may cause dropouts of data. Any condition that causes the tape to be lifted as little as 1/1000-inch from the read/write head will cause the signal to fall below effective sensitivity of the head. Although some of these particles may also become sufficiently embedded in the tape surface to resist cleaning and cause degradation, the best way to combat this contamination is to clean the tape drive every time the tape reel is changed. The heads should be cleaned by use of a cotton swab or lint-free pad moistened with Genesolve-D (an Allied Chemical Co. trademark), or Freon TF (a DuPont Co. trademark), or similar recommended cleaner; but never use carbon tetrachloride or vythene for cleaning magnetic tape. When cleaning the tape, the direction of wiping should be the same as that of the tape movement during normal operation.

A primary cause of contamination is from mounting dirty tapes on the tape drive. Many times, even new tapes need cleaning. For example, if the tape is touched, the oily finger prints form excellent holding surfaces for dust and lint.

Magnetic tapes may become contaminated by being stored in dusty or dirty canisters. To maintain a clean canister, it's cover must be replaced while tape is not present and never left open with handle down to collect dust. These containers should be cleaned before storing tape in them, and before bringing them into the work area. Only canisters having gaskets which properly seal should be used for storing tape.

Dirt may be deposited on tapes because of an unclean work area. The atmospheric pressure in the work area should be positive with respect to surrounding areas to reduce contaminants from coming into the area. Also, food and drink should be prohibited around tapes. Neither should smoking be allowed, where ashes cause more problems than the smoke. Never use an eraser to alter the identification of a tape label, for the particles could create a dirty tape. Other precautions for holding down contamination in the work area involve construction and maintenance activities in the area. Vinyl and rubber floor tile is preferred over asphalt tile, and any carpet and drapes should be lint-free and static-free. Ceilings and walls should be of materials which do not flake or dust. Floors should not be waxed, for traffic will abrade the wax causing fine dust to contaminate the room. Every 24 hours, use a damp mop or a cleaner with a long hose attachment so that the power unit and exhaust bag can be located outside the room. Never sweep or use a dry mop.

CARE AND HANDLING

Care should also be used in handling of magnetic tapes. Edge damage of tape may occur from misthreading or from the tape not being properly aligned on the transport. Broken flanges on reels, warped reels, write ring not being seated properly, or the reel not being properly seated may also cause edge damage to the tape. Also, when the tape is wound unevenly, the individual tape edges could protrude adjacent layers of tape, making the tape vulnerable to edge damage. Memorex says data may be only 9 mils from the edge of tape, so only slight edge damage could cause loss of data.

To protect against inadvertently damaging the tape, handle the hub of the reel and not the flanges. When mounting a reel of tape onto the transport, push on the hub rather than the flanges. Also, never stack reels of tape flat or on top of others. If a reel is dropped, check it for cracked flanges and for damaged tape edges. Also, avoid fingernails from going between flanges.

ENVIRONMENT

The environment of the work area should be held constant. Temperatures should be held to 70°F $\pm$ 5°F and relative humidity should be held to 40% $\pm$ 5%. Allow all tapes brought into the room to reach environmental equilibrium before being used. They should be left in their container to stabilize slowly. If a tape is subjected to extremely low temperature and used before it returns to normal environmental conditions, it may become distorted. If the tape is subjected to start and stop action, its layers may shift due to momentum, called cinching. This cinching can permanently damage the tape by causing a series of creases or folds in the slipped area. When operating, these wrinkles will repeatedly lift the tape from the head, causing dropouts.

Magnetic tapes are erased by magnetic fields, therefore keep recorded tapes away from strong fields. Although most magnetic fields are diminished to a point that no effect will be detected beyond three inches from its source, closeness to these sources should be prevented. Magnets used for securing notes to metal surfaces, permanent magnets, door latches, flashlights, key chains, etc. can cause data loss. Motors, generators, and transformers are hazards that can accidentally erase information.

International Business Machines says not to expose tape to magnetic fields of more than 50 oersteds. While most security devices are safe for magnetic tapes, some at airport gates, store entrances, etc. may exhibit amplitudes in the order of 100 oersteds and should be avoided. Even X-rays which present no danger to tapes, themselves, may contain a transformer which creates a magnetic field of this magnitude. Even some 'dirty' power supplies can present hazards of this type to tapes.

STORAGE

Before storage, secure the free end of a tape to the next wrap on the reel by taping with a vinyl strip, then place tape in canister and store it on edge in the upright position. If it is expected that it will be stored for a long time, the storage canister should be placed in a sealed plastic bag. Also, a reasonable degree of water resistance should be provided. Stored tapes should be exercised once a year to relieve internal pressures.

SUMMARY

All of the above practices are important for any magnetic tape user to observe. However, their degree of importance may vary with the type of usage. For instance, the practice of cleaning a tape drive when changing reels would not be practical for business applications when reels are changed several times daily.

Should these recommendations be grouped as to importance for tape transports used in General Electric's supervisory instrumentation, the following groups would result:

Most Important:

1. Clean tape drive periodically
2. DO NOT smoke or eat around tapes.
3. Check alignment of tape after threading on transport.
4. Never use an eraser to alter a tape label.
5. Hold environment constant.
6. Avoid magnetic devices.

Lesser Important:

1. Replace covers on canisters while using tape.
2. Have positive pressure in work area.
3. Use reel hub; not flange, for handling and mounting.
4. Store canisters on edge.
5. Secure free end of tape.

Least Important:

1. Clean canister before storing tape in it.
2. Use canisters with sealing gaskets.
3. Avoid touching tape with hands.
4. Do not wax floors in work area.
5. Vacuum or damp-mop frequently.
6. Exercise stored tapes.

CONCLUSION

The retention of magnetic tape is such that the data will be available as long as the supervisory instrumentation equipment needs it. If proper care is taken, one should expect these tapes to retain their data for more than fifty years.